

|                               |                               |                        |
|-------------------------------|-------------------------------|------------------------|
| <b>Engineering Physics II</b> | <b>Faculty of Engineering</b> | <b>Date: 3 /8/2025</b> |
| <b>B<sub>1</sub> MI/E</b>     | <b>First Test - Modeule B</b> |                        |
| <b>Name:</b>                  | <b>Reg. No.:</b>              |                        |

**Section I: Answer all the following by putting true (T) or false (F):** **6 Marks**

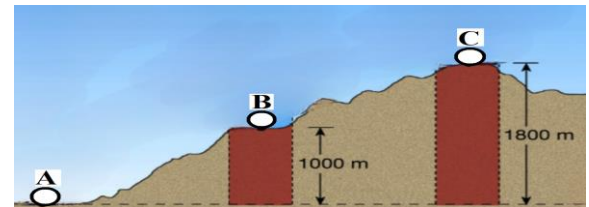
|                                                      |  |  |
|------------------------------------------------------|--|--|
| 1) The electric flux decreases as the area increases |  |  |
| 2) A suck pressure is a negative gauge pressure      |  |  |
| 3) A floating object is at rest                      |  |  |

**Section II: Answer all the following by selecting the correct choice:** **14 Marks**

1) If the two spheres come into contact and then separate. The charge on sphere B is:



|                                                |            |           |                      |                  |
|------------------------------------------------|------------|-----------|----------------------|------------------|
| a) 1.5 C                                       | b) 6 C     | c) 0      | d) 3 C               | e) None of these |
| 2) Which of the following has positive charge: |            |           |                      |                  |
| a) Electron                                    | b) Neutron | c) Proton | d) Electron & Proton | e) None of these |
| 3) From the figure shown below. The pressures  |            |           |                      |                  |



|                                                                                                                                                |                    |                 |                     |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|---------------------|------------------|
| a) $P_C = P_B$                                                                                                                                 | b) $P_A > P_B$     | c) $P_B < P_C$  | d) $P_A < P_C$      | e) None of these |
| 4) Partially immersed object with density equal to the liquid's density. The object will:                                                      |                    |                 |                     |                  |
| a) Floating                                                                                                                                    | b) Moving up       | c) Flying       | d) Sinking          | e) None of these |
| 5) 25 N object displaces 25 N of water. The object weight in water is:                                                                         |                    |                 |                     |                  |
| a) 10 N                                                                                                                                        | b) 25 N            | c) 15 N         | d) 0 N              | e) None of these |
| 6) The force 16 N. If a distance changes from 2 m into 8 m. The force is:                                                                      |                    |                 |                     |                  |
| a) 1 N                                                                                                                                         | b) 0.5 N           | c) 16 N         | d) 128 N            | e) None of these |
| 7) Object with $4\text{ m} \times 3\text{ m} \times 2\text{ m}$ having a density of $0.1\text{ kg/m}^3$ . The pressure in Pa on the ground is: |                    |                 |                     |                  |
| a) $20\text{ x}$                                                                                                                               | b) $0.83\text{ x}$ | c) $2\text{ x}$ | d) $0.083\text{ x}$ | e) None of these |

**Section III: Answer all the following by selecting the correct choice**

**Full and clear steps are required**

**20 Marks**

| Constant                                       |                                                     |                                      |
|------------------------------------------------|-----------------------------------------------------|--------------------------------------|
| Electron: $m = 9.11 \times 10^{-31}\text{ kg}$ | $K_e = 9 \times 10^9\text{ N.m}^2/\text{C}^2$       | Gravity: $g = 10\text{ m/s}^2$       |
| Proton: $m = 1.67 \times 10^{-27}\text{ kg}$   | $G = 6.67 \times 10^{-11}\text{ N.m}^2/\text{kg}^2$ | Water: $\rho_f = 1000\text{ kg/m}^3$ |

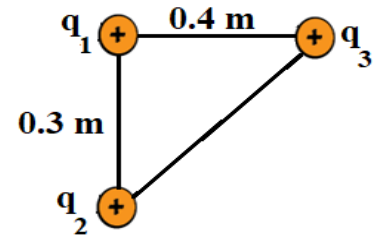
1) Two objects having  $m_1 = 2\text{ kg}$  and  $m_2 = 1\text{ kg}$ , separated by 30 cm. The gravitational force is:

|                                     |                                     |                                    |                                     |                  |
|-------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|------------------|
| a) $148.2 \times 10^{-11}\text{ N}$ | b) $44.47 \times 10^{-11}\text{ N}$ | c) $88.9 \times 10^{-11}\text{ N}$ | d) $296.4 \times 10^{-11}\text{ N}$ | e) None of these |
|-------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|------------------|

2) 300 N object in air displaces 120 N when it is fully immersed in a fluid of  $\rho_f = 1500 \text{ kg/m}^3$ . The volume of the metal is:

|                       |                        |                        |                        |                  |
|-----------------------|------------------------|------------------------|------------------------|------------------|
| a) $0.01 \text{ m}^3$ | b) $0.012 \text{ m}^3$ | c) $0.008 \text{ m}^3$ | d) $0.023 \text{ m}^3$ | e) None of these |
|-----------------------|------------------------|------------------------|------------------------|------------------|

2) The total electric force on  $q_3$  is:  $q_1 = 2 \text{ } \mu\text{C}$  ,  $q_2 = 2 \text{ } \mu\text{C}$  and  $q_3 = 3 \text{ } \mu\text{C}$



|           |           |           |           |                  |
|-----------|-----------|-----------|-----------|------------------|
| a) 0.46 N | b) 0.53 N | c) 0.37 N | d) 0.75 N | e) None of these |
|-----------|-----------|-----------|-----------|------------------|

4) Two opposite charges one is 4 times the other separated by distance  $r$ , if a ratio of charge to a distance  $r$  is  $2 \text{ } \mu\text{C/m}$ . The electric force on  $3q$  placed at distance  $r$  from a point where  $E = 0$  is:

|           |          |            |        |                  |
|-----------|----------|------------|--------|------------------|
| a) 1.33 N | b) 2.5 N | c) 0.075 N | d) 0 N | e) None of these |
|-----------|----------|------------|--------|------------------|

5) A room has dimensions of 3 m and 2 m and a height of 4 m. If the air density is  $1.21 \text{ kg/m}^3$ . The atmosphere force on the floor of the room is: Take  $P_{\text{atm}} = 10^5 \text{ Pa}$

|             |             |             |           |                  |
|-------------|-------------|-------------|-----------|------------------|
| a) 900000 N | b) 600000 N | c) 800000 N | d) 500000 | e) None of these |
|-------------|-------------|-------------|-----------|------------------|